

$$\#1. \vec{g} = [\sqrt{6}/6, \sqrt{6}/6, \sqrt{6}/6, \sqrt{2}/2]$$

$$\text{Let } \vec{g} = [\sqrt{6}/6, \sqrt{6}/6, \sqrt{6}/6]^T \text{ and } g_4 = \sqrt{2}/2.$$

$$\Rightarrow R = (g_4^2 - \vec{g}^T \vec{g})I + 2\vec{g}\vec{g}^T + 2g_4 S(\vec{g})$$

$$= (\frac{1}{2} - \cancel{3 \cdot \frac{1}{6}})I + 2 \times \frac{1}{6} \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix} + \frac{\sqrt{2}}{\sqrt{6}} \begin{bmatrix} 0 & -1 & 1 \\ 1 & 0 & -1 \\ -1 & 1 & 0 \end{bmatrix}$$

$$= \frac{1}{3} \begin{bmatrix} 1 & 1-\sqrt{3} & 1+\sqrt{3} \\ 1+\sqrt{3} & 1 & 1-\sqrt{3} \\ 1-\sqrt{3} & 1+\sqrt{3} & 1 \end{bmatrix}$$

$$\#2. \text{Let } \vec{g} = [g_1, g_2, g_3]^T, \text{ and } Q = [g^T, g_4]^T$$

$$\dot{\vec{g}} = \frac{1}{2} S(\omega) \vec{g} + \frac{1}{2} g_4 \omega. \quad \& \quad \dot{g}_4 = -\frac{1}{2} \omega^T \vec{g}. \quad \forall t \geq 0$$

$$\text{Claim: } Q^T Q = 1 \quad \forall \text{ time } t \geq 0$$

$$\dot{Q} = \begin{bmatrix} \frac{1}{2} S(\omega) & \frac{1}{2} \omega \\ -\frac{1}{2} \omega^T & 0 \end{bmatrix} Q.$$

$$\frac{d}{dt}(Q^T Q) = 2Q^T \dot{Q} = 2 \cdot \frac{1}{2} Q^T \begin{bmatrix} S(\omega) & \omega \\ -\omega^T & 0 \end{bmatrix} Q = Q^T \begin{bmatrix} S(\omega) & \omega \\ -\omega^T & 0 \end{bmatrix} Q$$

$$= [g_1 \ g_2 \ g_3 \ g_4] \begin{bmatrix} 0 & -\omega_z & \omega_y & \omega_x \\ \omega_z & 0 & -\omega_x & \omega_y \\ -\omega_y & \omega_x & 0 & \omega_z \\ -\omega_x & -\omega_y & -\omega_z & 0 \end{bmatrix} \begin{bmatrix} g_1 \\ g_2 \\ g_3 \\ g_4 \end{bmatrix}$$

$$= [g_1 \ g_2 \ g_3 \ g_4] \begin{bmatrix} -\omega_z g_2 + \omega_y g_3 + \omega_x g_4 \\ \omega_z g_1 - \omega_x g_3 + \omega_y g_4 \\ -\omega_y g_1 + \omega_x g_2 + \omega_z g_4 \\ -\omega_x g_1 - \omega_y g_2 - \omega_z g_3 \end{bmatrix}$$

$$= -\omega_z g_1 g_2 + \omega_y g_1 g_3 + \omega_x g_1 g_4 + \omega_z g_1 g_2 - \omega_x g_2 g_3 + \omega_y g_2 g_4 \\ - \omega_y g_1 g_3 + \omega_x g_2 g_3 + \omega_z g_3 g_4 - \omega_x g_1 g_4 - \omega_y g_2 g_4 - \omega_z g_3 g_4 \\ = 0.$$

$$\therefore Q^T Q \text{ is constant for any time } t \geq 0$$

$$\therefore Q^T Q = 1. \quad \forall \text{ time } t \geq 0$$

$$\#3. J = \text{diag}(200, 200, 200) \text{ kgm}^2, \text{ rest to rest maneuver. } \Rightarrow \vec{\omega}(0) = 0 = \vec{\omega}(t_f)$$

$$\vec{a} = [0.424, -0.566, 0.701]^T$$

$$R_0 = \exp[S(\vec{a})] = I + \sin \phi S(\vec{a}) + S(\vec{a})(1 - \cos \phi) \\ = I + \sin |S(\vec{a})| + (\vec{a}\vec{a}^T - I)(1 - \cos |S(\vec{a})|) \quad \rightarrow I = I_{3 \times 3} \\ = S(\vec{a}) \sin |S(\vec{a})| + \vec{a}\vec{a}^T (1 - \cos |S(\vec{a})|) + I \cos |S(\vec{a})|.$$

$$R_f = I_{3 \times 3}. \Rightarrow \phi_0 = 1 \quad \& \quad \phi_f = 0.$$

$$a. R = \exp(S(\vec{a})\phi(t))$$

$$\Rightarrow \dot{R} = S(\vec{a})\dot{\phi}(t) \exp(S(\vec{a})\phi(t)) = S(\vec{a})\dot{\phi}(t)R.$$

$$\text{Since } \dot{R} = S(\vec{\omega})R \text{ from rotation matrix kinematics,}$$

$$\Rightarrow S(\vec{\omega})R = S(\vec{a})\dot{\phi}(t)R$$

$$\Rightarrow \vec{\omega} = \vec{a}\dot{\phi}(t) \dots (B)$$

Assume that $\omega(t) = \begin{cases} -0.01 \text{ rad/s}^2 & \text{for } 0 \leq t \leq t_s \\ +0.01 \text{ rad/s}^2 & \text{for } t_s < t \leq t_f \end{cases}$

From Euler rotational EoM,

$$\begin{cases} I_x \dot{\omega}_x = (I_y - I_z) \omega_y \omega_z + M_x = M_x \\ I_y \dot{\omega}_y = (I_z - I_x) \omega_x \omega_z + M_y = M_y \\ I_z \dot{\omega}_z = (I_x - I_y) \omega_x \omega_y + M_z = M_z \end{cases} \Rightarrow M = J \ddot{\omega} = J \ddot{\alpha} \phi = J \ddot{\alpha} u \quad \dots (A)$$

$$\therefore \ddot{\omega}(t) = \ddot{\alpha} u(t)$$

$$\Rightarrow \omega(t) = \int_0^t \ddot{\alpha} u(\tau) d\tau$$

$$\text{If } 0 \leq t < t_s, \omega(t) = \int_0^t \ddot{\alpha} u(\tau) d\tau = \ddot{\alpha} \int_0^t -0.01 d\tau = -0.01 t \ddot{\alpha}$$

$$\text{If } t = t_s, \omega(t_s) = \int_0^{t_s} \ddot{\alpha} u(\tau) d\tau = -\ddot{\alpha} 0.01 t_s$$

$$\begin{aligned} \text{If } t_s < t \leq t_f, \omega(t) &= \int_0^t \ddot{\alpha} u(\tau) d\tau \\ &= \int_0^{t_s} \ddot{\alpha} u(\tau) d\tau + \int_{t_s}^t \ddot{\alpha} u(\tau) d\tau \\ &= -0.01 t_s \ddot{\alpha} + 0.01 (t - t_s) \ddot{\alpha} \\ &= 0.01 (2t_s - t) \ddot{\alpha} \end{aligned}$$

$$\Rightarrow \omega(t) = \begin{cases} -0.01 t \ddot{\alpha} & \text{for } 0 \leq t \leq t_s \text{ (rad/s)} \\ -0.01 (2t_s - t) \ddot{\alpha} & \text{for } t_s \leq t \leq t_f \text{ (rad/s)} \end{cases}$$

b. From (A), $M = J \ddot{\alpha} u = 200 \ddot{\alpha} u$

$$\Rightarrow M(t) = \begin{cases} -2 \ddot{\alpha} \text{ kgm}^2/\text{s}^2 & \text{for } 0 \leq t \leq t_s \\ +2 \ddot{\alpha} \text{ kgm}^2/\text{s}^2 & \text{for } t_s \leq t \leq t_f \end{cases}$$

c. Since the problem is rest to rest problem, $\ddot{\omega}(t_f) = 0$

$$\Rightarrow \ddot{\omega}(t_f) = -0.01 (2t_s - t_f) \ddot{\alpha} = 0$$

$$\Rightarrow 2t_s = t_f \therefore t_s = t_f / 2$$

From (B), $\ddot{\omega} = \ddot{\alpha} \phi(t)$

$$\ddot{\alpha} \phi(t) = \int_0^t \ddot{\alpha} \phi(\tau) d\tau + \ddot{\alpha} = \int_0^t \ddot{\omega}(\tau) d\tau + \ddot{\alpha}$$

If $0 \leq t < t_s = t_f/2$,

$$\ddot{\alpha} \phi(t) = \int_0^t -0.01 \tau \ddot{\alpha} d\tau + \ddot{\alpha} = \ddot{\alpha} (-0.005 t^2 + 1)$$

$$\Rightarrow \phi(t) = -0.005 t^2 + 1$$

$$\text{If } t = t_s, \phi(t_s) = -0.005 t_s^2 + 1$$

If $t_s < t \leq t_f$,

$$\begin{aligned} \ddot{\alpha} \phi(t) &= \int_0^{t_s} 0.01 \tau \ddot{\alpha} d\tau + \int_{t_s}^t -0.01 (2t_s - \tau) \ddot{\alpha} d\tau + \ddot{\alpha} \\ &= (0.005 t_s^2 - 0.02 t_s (t - t_s) + 0.005 (t^2 - t_s^2) + 1) \ddot{\alpha} \\ &= (0.01 t_s^2 - 0.02 t_s t + 0.005 t^2 + 1) \ddot{\alpha} \\ \Rightarrow \phi(t) &= 0.01 (t_s - t)^2 - 0.005 t^2 + 1 \end{aligned}$$

$$\text{Since } \phi(t_f) = 0, \phi(t_f) = -0.005 t_f^2 + 0.01 (t_s - t_f)^2 + 1 = 0$$

$$\Rightarrow t_f^2 = +400 \dots ??$$

$$\therefore t_f = 20$$

#4. Pitch axis control model. $J_y \ddot{\theta} = 3\omega_0^2 (-I_x + I_z) \theta + M_y$
 $\omega_0 = 0.0012 \text{ rad/sec}$
 $J = \text{diag}(150, 200, 100) \Rightarrow I_x = 150, I_y = 200, I_z = 100.$

test to test maneuver

$$\Delta\theta_0 = 45^\circ = \pi/4 \text{ rad} \quad \& \quad \Delta\theta_f = 0^\circ = 0 \text{ rad}$$

Let assume that this test to test maneuver can be performed using a single impulse, T .

Let t_s be a time that a single impulse is performed.

$$200 \ddot{\theta} = 3\omega_0^2 (-150 + 100) \theta + M_y$$

$$= -150\omega_0^2 \theta + M_y$$

$$\Rightarrow \ddot{\theta} = -(3/4)\omega_0^2 \theta + M_y/200$$

$$\Rightarrow \begin{bmatrix} \ddot{\theta} \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -(3/4)\omega_0^2 & 0 \end{bmatrix} \begin{bmatrix} \theta \\ \dot{\theta} \end{bmatrix} + \begin{bmatrix} 0 \\ 1/200 \end{bmatrix} M_y$$

If $t \in [0, t_s]$,

$$\begin{bmatrix} \theta(t) \\ \dot{\theta}(t) \end{bmatrix} = \exp \left(\begin{bmatrix} 0 & 1 \\ -(3/4)\omega_0^2 & 0 \end{bmatrix} t \right) \begin{bmatrix} \pi/4 \\ 0 \end{bmatrix}$$

$$\exp \left(\begin{bmatrix} 0 & 1 \\ -a & 0 \end{bmatrix} t \right) = I + At + \frac{1}{2!} (At)^2 + \dots \text{ where } a = \frac{3}{4}\omega_0^2$$

$$= I + At + \frac{1}{2!} (-at^2)I + \frac{1}{3!} (-at^3)A + \dots$$

$$= I \left(\sum_{i=0}^{\infty} \frac{(-at^2)^i}{(2i)!} \right) + A \left(\sum_{i=1}^{\infty} \frac{(-a)^i t^{2i-1}}{(2i-1)!} \right)$$

$$= \cos(\sqrt{a}t)I + \frac{1}{\sqrt{a}} \sin(\sqrt{a}t)A$$

$$= \begin{bmatrix} \cos \sqrt{a}t & \frac{1}{\sqrt{a}} \sin(\sqrt{a}t) \\ -\sqrt{a} \sin(\sqrt{a}t) & \cos \sqrt{a}t \end{bmatrix}$$

$$\begin{bmatrix} \theta(t) \\ \dot{\theta}(t) \end{bmatrix} = \begin{bmatrix} \cos \sqrt{a}t & \sin(\sqrt{a}t)/\sqrt{a} \\ -\sqrt{a} \sin \sqrt{a}t & \cos \sqrt{a}t \end{bmatrix} \begin{bmatrix} \pi/4 \\ 0 \end{bmatrix}$$

Thus at $t = t_s$, $\theta(t_s) = (\pi/4) \cos \sqrt{a}t_s = (\pi/4) \cos \sqrt{a}t_s$
 $\dot{\theta}(t_s) = -\sqrt{a} (\pi/4) \sin \sqrt{a}t_s$

At $t = t_s^+$, $\theta(t_s^+) = \theta(t_s)$ & $\dot{\theta}(t_s^+) = \dot{\theta}(t_s) + T$.

If $t \in (t_s, t_f)$,

$$\begin{bmatrix} \theta(t) \\ \dot{\theta}(t) \end{bmatrix} = \begin{bmatrix} \cos \sqrt{a}(t-t_s) & \sin \sqrt{a}(t-t_s)/\sqrt{a} \\ -\sqrt{a} \sin \sqrt{a}(t-t_s) & \cos \sqrt{a}(t-t_s) \end{bmatrix} \begin{bmatrix} (\pi/4) \cos \sqrt{a}t_s \\ -\sqrt{a} (\pi/4) \sin \sqrt{a}t_s + T \end{bmatrix}$$

At $t = t_f$,

$$\begin{bmatrix} \theta(t_f) \\ \dot{\theta}(t_f) \end{bmatrix} = \begin{bmatrix} \cos \sqrt{a}(t_f-t_s) & \sin \sqrt{a}(t_f-t_s)/\sqrt{a} \\ -\sqrt{a} \sin \sqrt{a}(t_f-t_s) & \cos \sqrt{a}(t_f-t_s) \end{bmatrix} \begin{bmatrix} (\pi/4) \cos \sqrt{a}t_s \\ -\sqrt{a} (\pi/4) \sin \sqrt{a}t_s + T \end{bmatrix}$$

Since $\cos^2 \sqrt{a}(t_f-t_s) + \sin^2 \sqrt{a}(t_f-t_s) = 1$, STM is invertible $\forall t_s, t_f$.

It means that there exist only one element in $\ker(\text{STM})$ for any t_s and t_f .
 $= [0, 0]^T$

Thus, $\cos \sqrt{a}t_s = 0$ and $-\sqrt{a} (\pi/4) \sin \sqrt{a}t_s + T = 0$.

If $\sqrt{a}t_s = n\pi + \frac{\pi}{2}$, then $\sin \sqrt{a}t_s = 1$.

$$\Rightarrow T = \sqrt{a}(\pi/4) \approx 8.1621e-04 \text{ rad/sec.}$$

If $\sqrt{a}t_s = n\pi + 3\pi/2$, then $\sin \sqrt{a}t_s = -1$.

$$\Rightarrow T = -\sqrt{a}(\pi/4) \approx -8.1621e-04 \text{ rad/sec}$$

The smallest possible t_s is 1511.4995 sec and $T = 8.1621e-04 \text{ rad/s}$

Thus control moment $M_y = 200T = 0.1632 \text{ kgm}^2/\text{sec}^2$.

Hence, if transfer time t_f is greater than 1511.4995 sec, then rest to rest maneuver can be performed using a single impulse.

$\Rightarrow$ The claim is true if $t_f \geq 1511.4995 \text{ sec}$.

The times instant the impulse should applied are $(2n-1)\pi/(2\sqrt{a})$ for any integer $n \in \mathbb{Z}$, and this times are same as the transfer time.

#5. $I_y \ddot{\Delta\theta} = 3\omega_0^2(-I_x + I_z) \Delta\theta + M_y$.

$$\omega_0 = 0.0012 \text{ rad/s. } J = \text{diag}(150, 100, 200) \text{ kgm}^2.$$

a. $M_y = 0 \Rightarrow 100 \ddot{\Delta\theta} = +150\omega_0^2 \Delta\theta$.

$$\Rightarrow \ddot{\Delta\theta} = + (3/2) \omega_0^2 \Delta\theta$$

$$\Rightarrow \begin{bmatrix} \dot{\Delta\theta}(t) \\ \ddot{\Delta\theta}(t) \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ + (3/2) \omega_0^2 & 0 \end{bmatrix} \begin{bmatrix} \Delta\theta \\ \dot{\Delta\theta} \end{bmatrix}$$

$$\text{eigenvalues: } \lambda^2 - \frac{3}{2} \omega_0^2 = 0 \Rightarrow \lambda = \pm \sqrt{\frac{3}{2}} \omega_0$$

Since there is positive real eigenvalue, then this system is NOT stable whatever an initial state is.

Hence the pitch equilibrium $(\Delta\theta, \dot{\Delta\theta}) = (0, 0)$ is unstable.

b. Let $M_y = -k_p \Delta\theta - k_d \dot{\Delta\theta}$

$$\Rightarrow 100 \ddot{\Delta\theta} = 150\omega_0^2 \Delta\theta - k_p \Delta\theta - k_d \dot{\Delta\theta}$$

$$\Rightarrow \ddot{\Delta\theta} + 0.01k_d \dot{\Delta\theta} + (0.01k_p - 1.5\omega_0^2) \Delta\theta = 0$$

$$\Rightarrow \{s^2 + 0.01k_d s + (0.01k_p - 1.5\omega_0^2)\} \Delta\theta_s = 0$$

$$\Rightarrow s^2 + 0.01k_d s + (0.01k_p - 1.5\omega_0^2) = s^2 + 0.0112s + 0.000064$$

$$\Rightarrow 0.01k_d = 0.0112 \quad \& \quad 0.01k_p - 1.5\omega_0^2 = 0.000064$$

$$\Rightarrow k_d = 1.12 \quad \& \quad k_p = 6.616e-03$$

Thus when $M_y = -6.616e-03 \Delta\theta - 1.12 \dot{\Delta\theta}$, the system is stable.